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## Preface

Lec 1 09-08 Tuesday

### NO LATES WILL BE ACCEPTED

“work hard on your assignments”

“the test os easier than the HW because you only get an hour”

“abstraction gap”

158/240 throw it into the deep end

“very little computation in this class”

*write proofs in prose more than symbols*

4-5 hours outside of class

lec (tue, thu) → pset drops (fri) → tut (following thu) → pset due (sun 8pm)

## I Basics

### 1 Sets

Lec 1 09-08 Tuesday

**Definition 1.1.** A set is a collection of elements. An element  $x$  is in the set  $S$ , we write  $x \in S$ . If the element  $x$  is not in the set  $S$ , we write  $x \notin S$ .

**Example 1.1.1.** Let  $P = \{x : x \text{ is a prime number}\}$ .

**Definition 1.2.** Two sets  $S$  and  $T$  are equal if they contain the same elements. In that case we write  $S = T$ .

A set can be defined either by listing the elements in the set, or by giving a characterization of these elements.

$$A = \{x : x \text{ is a positive integer dividing } 6\}$$

$$A = \{1, 2, 3, 6\} = \{2, 6, 1, 3\}$$

**Definition 1.3.** If  $S$  and  $T$  are sets such that all elements of  $S$  are contained in  $T$ , we say that  $S$  is a subset of  $T$  and write  $S \subseteq T$ . If  $S$  is a subset of  $T$  and is not equal to  $T$ , we say that  $S$  is a proper subset of  $T$  and write  $S \subsetneq T$ .

**Definition 1.4.** The empty set is the set with no elements, denoted  $\emptyset$ .

**Definition 1.5.** If  $S$  and  $T$  are sets, we have two operations

- $S \cup T = \{x : x \in S \text{ or } T \text{ or both}\}$  is called the union of  $S$  and  $T$ .
- $S \cap T = \{x : x \in S \text{ and } T\}$  is called the intersection of  $S$  and  $T$ .

**Example 1.5.1.** Let  $\mathbb{N} = \{x \in \mathbb{Z} : x > 0\} = \{1, 2, 3, 4, \dots\}$  be the set of natural numbers, and let  $-\mathbb{N} = \{x \in \mathbb{Z} : x < 0\}$  be the set of negative integers. Then  $\mathbb{N} \cup -\mathbb{N} = \{x \in \mathbb{Z} : x \neq 0\}$ .  $\mathbb{N} \cap \mathbb{N} = \emptyset$

**Definition 1.6.** Let  $S_1, \dots, S_k$  sets. Then,

$$\bigcup_{i=1}^k S_i = \{x : x \in S_i \text{ for at least one } 1 \leq i \leq k\}$$

is the union of sets  $S_i$ . Likewise,

$$\bigcap_{i=1}^k S_i = \{x : x \in S_i \forall 1 \leq i \leq k\}$$

is the intersection of sets  $S_i$ .

**Definition 1.7.** Let  $S$  and  $T$  be sets, The **Cartesian product** is defined by  $S \times T = \{(s, t) : s \in S, t \in T\}$ .

## 2 Functions

**Definition 2.1.** Let  $S$  and  $T$  be non-empty sets. A function  $f$  from  $S$  to  $T$ , denoted  $f : S \rightarrow T$ , is a rule assigning to each  $x \in S$  an unique element  $f(x) \in T$ .

In that case, we say that

- $f(x)$  is the **image** of  $x$  under  $f$ .
- $x$  is the **preimage** of  $f(x)$  under  $f$ .
- $S$  is the **domain** of  $f$ .
- $T$  is the **codomain** of  $f$ .

- the **range** of  $f$  is the set  $f(x) : x \in S$ .
- if  $U \subseteq S$ , then the image of  $U$  is  $f(U) = \{f(x) : x \in U\}$ .
- if  $V \subseteq T$ , then the preimage of  $V$  is  $f^{-1}(V) = \{x : f(x) \in V\}$ .

**Definition 2.2.** Two functions  $f, g : S \rightarrow T$  are equal if  $f(x) = g(x), \forall x \in S$ .

**Definition 2.3.** If every element in the range of  $f$  has a unique preimage, we say  $f$  is **injective**, or an **injection** (one-to-one).

**Definition 2.4.** If every element in the codomain is in the range, we say  $f$  is **surjective**, or a **surjection** (onto).

**Definition 2.5.** If  $f$  is both injective and surjective, we say  $f$  is **bijective**, or a **bijection**.

**Example 2.5.1.** Let  $f : \mathbb{Z} \rightarrow \mathbb{N}$  be defined by  $f(x) = |x| + 1$ . Then  $f$  is surjective but not injective.

**Definition 2.6.** If  $f : S \rightarrow T$  and  $U \subseteq S$ , then the restriction of  $f$  to  $U$ , denoted  $f|_U$ , is defined by  $f|_U(x) = f(x)$  for all  $x \in U$ .

**Definition 2.7.** If  $f : S \rightarrow T$  and  $g : T \rightarrow U$  are functions, then the function which sends  $x \in S$  to  $g(f(x))$  is the composite  $g \circ f : S \rightarrow U$ , given  $g \circ f(x) = g(f(x))$ . Note even if  $f, g : S \rightarrow S$ , we don't necessarily have  $f \circ g = g \circ f$ .

**Definition 2.8.** A function  $f : S \rightarrow T$  is **invertible** if there exists a function  $g : T \rightarrow S$  such that  $g \circ f(x) = x, \forall x \in S$ , and  $f \circ g(y) = y, \forall y \in T$ . If it exists, such a  $g$  is the **inverse** of  $f$  and is denoted  $f^{-1}$ .

### 3 Fields

Lec 2 09-10 Thursday

**Definition 3.1.** A **field**  $F$  is a set with 2 operations, **note that these operations are closed**:

- $+: F \times F \rightarrow F$  (**sum**)
- $\cdot: F \times F \rightarrow F$  (**product**)

that obeys the following axioms:

(F1)  $a + b = b + a$  and  $a \cdot b = b \cdot a$  (**commutativity**)

(F2)  $(a + b) + c = a + (b + c)$  and  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$  (**associativity**)

(F3)  $\exists$  distinct elements  $0, 1 \in F$  s.t.  $0 + a = a$  and  $1 \cdot a = a$  for all  $a \in F$ . (**existence of identities**)

(F4)  $\forall a \in F$  and  $\forall b \in F, b \neq 0$ , there exists  $c, d \in F$  s.t.  $a + c = 0$  and  $b \cdot d = 1$ . (**existence of inverses**)

(F5)  $\forall a, b, c \in F$ , then  $a \cdot (b + c) = a \cdot b + a \cdot c$ . (**distributivity of  $\cdot$  over  $+$** )

**Example 3.1.1.** Some fields:

- $(\mathbb{R}, +, \cdot)$
- $(\mathbb{Q}, +, \cdot)$
- $(\mathbb{Z}_2 = \{0, 1\}, +, \cdot)$

[!NOTE] memorize these axioms

[!NOTE] someone said  $\mathbb{Z}/p$  lmao

[!NOTE] Can you redefine  $+$  and  $\cdot$  on  $\mathbb{Z}$  to make it a field? Yes!

**Theorem 3.2** (Cancellation Result). *Let  $F$  be a field and  $a, b, c \in F$ . Then: - if  $a + c = b + c$ , then  $a = b$  - if  $a \cdot c = b \cdot c$ , and  $c \neq 0$ , then  $a = b$*

*Proof.* Let  $d \in F$  s.t.  $c \cdot d = 1$ .

$$(a \cdot c) \cdot d = (b \cdot c) \cdot d \quad (3.1)$$

$$a \cdot (c \cdot d) = b \cdot (c \cdot d) \quad (3.2)$$

$$a \cdot 1 = b \cdot 1 \quad (3.3)$$

$$a = b \quad (3.4)$$

A similar proof exists for the first statement. ■

**Corollary 3.2.1.** *The elements  $0, 1 \in F$  are unique. The inverses  $c, d \in F$  are also unique.*

[!NOTE] something about ring idempotent

## 4 The complex numbers

**Definition 4.1.** Let the complex numbers  $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$ , equipped with the following operations:

- $+_C : \mathbb{C} \rightarrow \mathbb{C}$  given by  $(a + bi) +_C (c + di) = (a + c) + (b + d)i$
- $\cdot_C : \mathbb{C} \rightarrow \mathbb{C}$  given by  $(a + bi) \cdot_C (c + di) = (ac - bd) + (ad + bc)i$ , with  $i^2 = -1$  Elements of  $\mathbb{C}$  are called the **complex numbers**, elements of the form  $0 + bi$  are called the **imaginary numbers**.

**Example 4.1.1.**  $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ , naturally since  $\mathbb{R}$  is just all the elements of the form  $a + bi$ .

**Definition 4.2.** Any complex number  $z = a + bi$ :

- has a **modulus** (absolute value)  $|a + bi| = \sqrt{a^2 + b^2}$ .
- has a **conjugate**  $\bar{z} = a - bi$ .

**Theorem 4.3.**  $\mathbb{C}$  is a field.

*Proof.* The proof is left as an exercise to the reader. ■

## 5 Equivalence relations

Lec 3 09-15 Tuesday

[!quote] “I as an algebraist think they are the most important concept in algebra.”

**Definition 5.1.** Let  $S$  be a set. A relation  $R$  is a subset  $R \subseteq S \times S$ . We say that a pair  $(x, y) \in S$  satisfies  $R$  if  $(x, y) \in R$ . We denote  $(x, y) \in R$  by  $x \sim y$ .

**Definition 5.2.** A relation  $\sim$  on a set  $S$  is a equivalence relation if it satisfies the following properties  $\forall s, t, u \in S$ :

- $s \sim s$  (\*\*reflexivity\*\*)
- $s \sim t \implies t \sim s$  (\*\*symmetry\*\*)
- $s \sim t$  and  $t \sim u \implies s \sim u$  (\*\*transitivity\*\*)

*Exercise.* Consider  $w \in W$ , the set of all english words. What equivalence classes exist on  $W$ ?

**Example 5.2.1.** Some provided by people:

- $w_1 \sim w_2$ , if they start with the same letter
- $w_1 \sim w_2$ , if they start with the same number of characters
- $w_1 \sim w_2$ , if they are the same word - \*\*for any set you get for free a equivalence relation (equality)\*\* - and also a trivial equivalence relation  $s \sim_{tr} t, \forall s, t \in S$

**Definition 5.3.** If  $(S, \sim)$  is a set equipped with a equivalence relation, and  $t \in S$ . Let \*\*equivalence class\*\*  $C_t = \{s \in S : s \sim t\}$ .

**Theorem 5.4.** For  $(S, \sim)$ , there is a subset  $T \subseteq S$  that satisfies the following:

- $\bigcup_{t \in T} C_t = S$
- If  $t, t' \in T$  and  $t \neq t'$ , then  $C_t \cap C_{t'} = \emptyset$ .

$T$  is a \*\*set of representative\*\* of  $S$  for equivalence relation  $\sim$ . Note that this implies  $C_t$  partitions  $S$ , since they are all pairwise disjoint.

## 6 Modular arithmetic

Lec 3 09-15 Tuesday

**(1) Def.\*** Let  $m \in \mathbb{N}$ , and  $a, b \in \mathbb{Z}$ . We say that  $a$  is congruent to  $b$  mod  $m$  denoted,

$$a \equiv b \pmod{m}$$

or,

$$a \equiv b \pmod{m}$$

if  $m | (a - b)$ . **Ex.\*** -  $11 \equiv 18 \pmod{7}$  -  $0 \equiv 7 \pmod{7}$  **Exercise.\*** What  $m$  makes  $3 \equiv 45 \pmod{m}$  true? -  $m = 1, 2, 3, 6, 7, 14, 21, 42$

**(2)** **Lemma.** Let  $m \in \mathbb{N}$ , and  $a, b \in \mathbb{Z}$ . Then,  $a \equiv b \pmod{m}$  iff there exists  $k \in \mathbb{Z}$  s.t.  $a = b + km$ . **Proof.** **FILL IN FROM NOTES, pretty intuitive**

**(3)** **Thm.** (**Division with remainder**) Let  $m \in \mathbb{N}$ . For each  $a \in \mathbb{Z}$ , there exists a unique  $r \in \{0, 1, \dots, m-1\}$  such that  $a \equiv r \pmod{m}$ . **Proof.** We will show that there exist unique integers  $k$  and  $r$  such that  $a = km + r$  by **Lemma 2**, this give  $a \equiv r \pmod{m}$ . **Existence.** Consider all the integer multiples of  $m$ :

$$m\mathbb{Z} = \{0, \pm m, \pm 2m, \pm 3m, \dots\}.$$

These integers occupy the entire real line by the Archimedean property of the real numbers. [\[\[09-15 Equivalence relations and modular arithmetic-20260915201512028.webp\]\]](#) The integer  $a$  lies somewhere on the real line, in one of these intervals. Thus, it satisfies

$$km \leq a < (k+1)m$$

for some value  $k$ . Subtract  $km$  to obtain

$$0 \leq a - km < m.$$

Let  $r = a - km$ . Then  $r$  satisfies the desired condition. **Uniqueness.** Suppose that  $a = r + km = r' + k'm$ . Then

$$r - r' = (k' - k)m.$$

It follows that  $r - r'$  is a multiple of  $m$ . But since  $0 \leq r, r' < m$ , we have  $-m < r - r' < m$ . The only multiple of  $m$  on this interval is 0. So  $r - r' = 0 \implies r = r'$  and hence  $k = k'$ .

As a consequence, we can deduce that

$$\mathbb{Z} = \bigcup_{i=0}^{m-1} C_i$$

We call  $\{0, \dots, m-1\}$  the **standard representatives** of their respective **congruence classes**  $C_i$  for congruence mod  $m$ .

**(4)** **Thm.** (**Addition and multiplication of congruence classes**) If  $a \equiv b \pmod{m}$  and  $c \equiv d \pmod{m}$ , then

$$a + c \equiv b + d \pmod{m}, \quad a \cdot c \equiv b \cdot d \pmod{m}.$$

**Proof.** see notes,

**(5)** **Def.** The **integers modulo  $m$  (or mod  $m$ )**, denoted  $\mathbb{Z}/m\mathbb{Z}$ , is the set of congruence classes mod  $m$ . By the previous theorem,  $\mathbb{Z}/m\mathbb{Z}$  is equipped with addition and multiplication. In practice, we will denote the congruence classes in  $\mathbb{Z}/m\mathbb{Z}$  to be

$$\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{m-1}\}.$$

Let  $m = 4$ . Consider  $\mathbb{Z}/4\mathbb{Z}$ :

$$\bar{0} \cdot \bar{2} = \bar{0} \quad \bar{1} \cdot \bar{2} = \bar{2} \quad \bar{2} \cdot \bar{2} = \bar{0} \quad \bar{3} \cdot \bar{2} = \bar{2}$$

$\bar{2}$  does not have a multiplicative inverse!!!! So  $\mathbb{Z}/4\mathbb{Z}$  is not a field. Note that however  $\bar{3}$  does have a multiplicative inverse, it is only for (non-unity) divisors of  $m$  that lack inverses.

**(6)** **Thm.**  $\mathbb{Z}/m\mathbb{Z}$  is a field if and only if  $m$  prime number. Assume the field axioms except the existence of multiplicative inverses. Assume that if  $m$  is not prime then there exists

## 7 Finite fields on $\mathbb{Z}/m\mathbb{Z}$

Lec 4 09-17 Thursday

Go to tutorial!!! HW1 is due sunday at 8pm. Mathematics Support Initiative, see Quercus, math help club.

Recap. Recall that  $\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \dots, \overline{m-1}\}$  is the set of equivalence classes mod  $m$ , equipped with the operations  $+$  and  $\cdot$ , given by:

1. Do the operation as usual
2. Then, take the remainder

*Remark.* What is a prime number?

**Definition 7.1.** Let  $p \in \mathbb{N}$ ,  $p > 1$ , then  $p$  is prime if  $\forall x, y \in \mathbb{Z}$ ,

$$p \nmid x \text{ and } p \nmid y \implies p \nmid xy$$

asfasf

*Remark.* What is a prime number?

Check this as an exercise! This is also called Euclid's lemma.

**Theorem 7.2.** Let  $m \in \mathbb{N}$ . If  $m$  is prime, then  $\mathbb{Z}/m\mathbb{Z}$  is a field. blahblahblahblahblahblah  
blah blah blah blah

*Remark.* What is a prime number?

*Proof.* First, check the field axioms, with (F4) holding for addition. ■

## II Vector spaces

### 8 Vector spaces and their properties

Lec 5 09-22 Tuesday